USER INTERFACE DESIGN

EX No: 10 **240701463**

Sales Data Visualization using JavaScript

AIM:

The aim is to create data visualizations such as pie charts and bar graphs to analyze sales data using JavaScript.

PROCEDURE:

Step 1: Create HTML File (index.html)

Create an HTML file to display the charts.
Include canvas elements and the Chart.js library.

Step 2: Create JavaScript File (script.js)

Create a JavaScript file to define the sales data and generate charts.

Step 3: Generate Charts

Use Chart.js to create:

- Pie Chart for sales distribution
- Bar Chart for category-wise comparison

SOURCE CODE:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sales Data Visualization</title>
```

```
    <style>
        body {
            font-family: Arial;
            text-align: center;
            margin: 50px;
        }
        canvas {
            margin: 20px auto;
        }
    </style>
</head>
<body>
    <h1>Sales Data System</h1>
    <canvas id="pieChart" width="400" height="400"></canvas>
    <canvas id="barChart" width="400" height="400"></canvas>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <script src="script.js"></script>
</body>
</html>
```

---

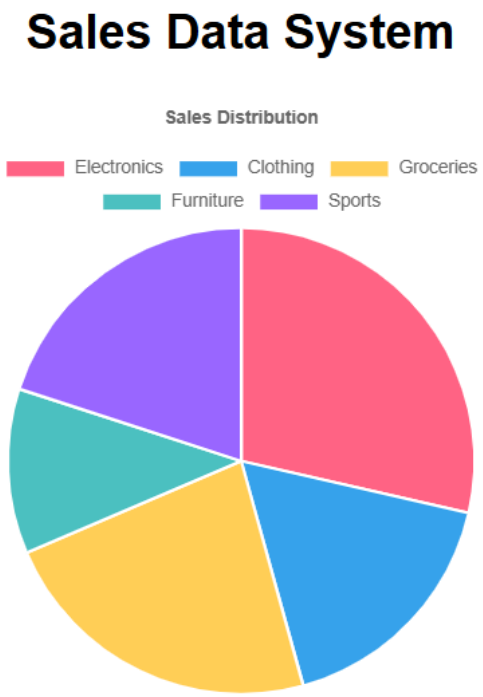
script.js

```
const salesData = {
    labels: ['Electronics', 'Clothing', 'Groceries', 'Furniture', 'Sports'],
    datasets: [{
        label: 'Sales',
        data: [500, 300, 400, 200, 350],
        backgroundColor: [
```

```
            '#FF6384',
            '#36A2EB',
            '#FFCE56',
            '#4BC0C0',
            '#9966FF'
        ]
    }]
};
const ctxPie = document.getElementById('pieChart').getContext('2d');
new Chart(ctxPie, {
    type: 'pie',
    data: salesData,
    options: {
        responsive: true,
        plugins: {
            title: {
                display: true,
                text: 'Sales Distribution'
            }
        }
    }
});
const ctxBar = document.getElementById('barChart').getContext('2d');
new Chart(ctxBar, {
    type: 'bar',
    data: salesData,
    options: {
        responsive: true,
```

```
      plugins: {
        title: {
          display: true,
          text: 'Sales by Category'
        }
      },
      scales: {
        y: {
          beginAtZero: true
        }
      }
    }
});
```

OUTPUT:



Pie Chart showing sales distribution

Bar Chart showing sales by category

RESULT:

Thus, the sales data visualization system was successfully created using JavaScript and Chart.js.